

CHRONON-1

Progressive Validation of a Local Tempo Field

Experimental protocol, falsifiability and no-go conditions

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CHRONON-1 — Statistical Analysis Plan (SAP) (*Br  cheteau, 2025; DOI 10.5281/zenodo.18072280*),
CHRONON-1 — Experimental of the Chronon Field Residual (*Br  cheteau, 2025; DOI 10.5281/zenodo.17571491*),
and CHRONON-1 v1.0.0 — Official Software Release (*Br  cheteau, 2025; DOI 10.5281/zenodo.17842188*).

Abstract

We present a falsifiable experimental testing program for a non-metric local tempo field, denoted $\Phi(x)$, constructed so as to fully preserve General Relativity and the structure of light cones. The protocol relies exclusively on differential observables, after complete subtraction of known relativistic predictions.

The approach is structured as a progressive validation in five stages, ranging from methodological credibility (public data, frozen pipeline, deterministic replication) to the search for a coherent network signature at the multi-site scale. The primary observable is a signed residual slope ε_Φ in vertical optical clock comparisons, expressed as a function of gravitational exposure $X = g\Delta h/c^2$. Orthogonal confirmatory channels based on quantum coherence (T_2) and network spectral analyses are integrated, without ever being able to rescue a primary failure.

The protocol is fully preregistered: a single primary endpoint, a single decision statistical model, explicit anti-systematic logical gates, and thresholds fixed prior to any analysis. Possible outcomes are deliberately restricted to three categories: robust detection, quantitative constraint, or formal demonstration of non-testability (no-go) within the considered framework.

This document does not aim to predict an outcome, but to guarantee that any obtained result — positive or negative — has a clear epistemic value, reproducible and irreversible.

Keywords: chronon field, time metrology, optical clocks, experimental falsifiability, quantum coherence, differential tests, general relativity

OPERATIONAL SUMMARY

Goal (in one sentence). Test, in an *audit-proof* manner, the existence of a non-metric residual compatible with a local tempo field $\Phi(x)$, *after complete subtraction* of known relativistic predictions.

Primary endpoint (2 lines).

- **Observable:** $\hat{\varepsilon}_\Phi$ (signed residual slope, dimensionless).
- **Regression:** *clock residual* Y vs *exposure* X : $Y = \alpha + \hat{\varepsilon}_\Phi X + \epsilon$.

Minimal definitions.

$$X \equiv \frac{g \Delta h}{c^2}, \quad Y \equiv y_{\text{mes}} - y_{\text{GR}} \quad (\text{after fixed GR corrections}).$$

Decision (3 cases, fixed thresholds).

Outcome	Condition (all required)
PASS–SIGNAL	All gates G1–G6 = PASS <i>and</i> $ t \geq 5$ with $t \equiv \hat{\varepsilon}_\Phi / \text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi)$
PASS–LIMIT	All gates G1–G6 = PASS <i>and</i> $ t < 5 \Rightarrow 95\%$ CL bound on $ \varepsilon_\Phi $
FAIL–NO-GO	At least one gate = FAIL $\Rightarrow$ <i>non-interpretable</i> (no physical conclusion)

Gates G1–G6 (one-line summary each).

Gate	Input	Test	PASS / FAIL (summary)
G1	control $\Delta h \approx 0$	same regression	PASS if $ t_0 < t^*$; FAIL otherwise
G2	swap A $\leftrightarrow$ B	slope comparison	PASS if sign/amplitude compatible; FAIL otherwise
G3	link permutations	subsets	PASS if no link dependence; FAIL otherwise
G4	randomized schedule	temporal permutations	PASS if stability of $\hat{\varepsilon}_\Phi$; FAIL otherwise
G5	environmental covariates	partial regression / veto	PASS if covariates do not “carry” the signal; FAIL otherwise
G6	robustness	LOO / bootstrap / outliers	PASS if slope is stable; FAIL otherwise

Mandatory outputs (short replication pack).

- **Figures:** (i) Y vs X with fit + CI; (ii) PASS/FAIL gate dashboard; (iii) residual diagnostics.
- **Tables:** (i) $\hat{\varepsilon}_\Phi \pm 1\sigma + 95\%$ CI; (ii) gate table (stat, p-val/criterion, PASS/FAIL); (iii) frozen parameters.
- **Logs:** metadata (clocks, links, geodesy, environment), exclusions (if allowed) *with justification*.
- **Reproducibility:** scripts + config + version (git) + **checksum** (SHA256) of data and outputs.

Survival rule. Secondary analyses serve only to *break* a false signal or to *document* a limit; they can never turn a FAIL into a PASS.

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1. MOTIVATION, STAKES AND TESTED HYPOTHESIS

1.1. Physical problem

The **CHRONON** project aims to test the existence of a local tempo field $\Phi(x)$, without modifying relativistic geometry or light cones. General Relativity (GR) is assumed exact for the metric; any deviation sought is *strictly differential* and appears only after complete subtraction of the known relativistic gravitational effect.

Optical clocks, through their extreme sensitivity to the gravitational potential, are used as *probes of local temporal coherence*. After GR correction, any residual correlated with gravitational exposure is either:

- an instrumental or environmental artifact,
- or the signature of a new non-metric physical structure.

1.2. Central risk: metrological self-deception

At precision levels on the order of 10^{-18} to 10^{-19} , an apparent signal can emerge from:

- slow temporal drifts correlated with the protocol,
- asymmetric instrumental biases between clocks,
- link artifacts (fiber, GNSS, free-space),
- uncontrolled environmental gradients,
- geodetic errors on Δh .

Any claim of new physics therefore requires a process whose logical rigor exceeds the instrumental sensitivity itself. CHRONON adopts a **falsifiability-first** approach: each step is designed to invalidate the signal hypothesis before attempting to confirm it.

1.3. Single observable and hypotheses

After complete subtraction of the relativistic prediction (GR), we define the primary observable and its exposure variable in a **operational** way:

$$Y \equiv \delta y_{\text{res}} = y_{\text{mes}} - y_{\text{GR}}, \quad X \equiv \frac{\Delta U}{c^2}.$$

Here ΔU is the gravitational potential difference (high – low) used in the GR correction. In the weak and local regime where the protocol is executed, we use the approximation:

$$\Delta U \simeq g \Delta h \quad \Rightarrow \quad X \simeq \frac{g \Delta h}{c^2},$$

with g locally documented and Δh geodetically traceable.

Sign convention (frozen). We set $\Delta h > 0$ for an elevation (“high” site above the “low” site), so $\Delta U > 0$ and $X > 0$ by convention. The physical sign of the residual *is not presupposed*, but the parameter reading is frozen:

$$Y = \alpha + \varepsilon_{\Phi} X + \epsilon,$$

where ε_{Φ} is dimensionless.

- $\varepsilon_{\Phi} > 0$ means: *residual* Y increasing with exposure X (thus, in this protocol, with altitude).
- $\varepsilon_{\Phi} < 0$ means: *residual* Y decreasing with X .

Null hypothesis (H0).

$$\varepsilon_{\Phi} = 0.$$

General Relativity is sufficient; any residual is compatible with noise and controlled systematics.

Signal hypothesis (H1).

$$\varepsilon_{\Phi} \neq 0.$$

A persistent linear residual, invariant under permutations and controls, indicates a non-metric coupling compatible with a tempo field $\Phi(x)$.

Minimal hypotheses (necessary) — closed list. The verdict on ε_Φ is admissible only if the following minimal conditions are satisfied:

1. **Minimal stationarity:** within each analysis window, the dominant statistical properties of the noise do not drift in a way correlated with X (no drift confounded with altitude).
2. **Quality of ΔU :** ΔU (or its proxy $g\Delta h$) is traceable, with documented uncertainties; applied geophysical corrections are *pre-fixed* (no optimization on residuals).
3. **Independence of dominant noises:** the dominant instrumental/environmental contributions do not carry an unmodeled systematic dependence on X (otherwise *gate FAIL*).
4. **Metrological comparability:** comparisons (clocks, links) are performed under a frozen and reproducible pipeline; the definition of Y is identical for all blocks.
5. **Absence of opportunistic selection:** no sub-selection of points, windows, or sites can be motivated by the observed value of $\hat{\varepsilon}_\Phi$.

Forbidden (anti-rewriting rules). After seeing the results, it is **forbidden** to:

- change the primary observable (Y) or the definition of exposure (X),
- modify the primary model or its decision thresholds,
- introduce a new “necessary” correction because it improves the signal,
- reclassify a secondary analysis as primary evidence.

Only one exception is allowed: a **documented erratum** (bug, unit, transcription error), published with traceability, without re-optimization and with full reproduction.

1.4. Guiding principle

CHRONON does not seek a positive result at any cost. It seeks a **robust verdict**:

- either a signal surviving all falsification tests,
- or a defensible quantitative limit,
- or the explicit identification of a blocking systematic.

Anti-self-deception rules (operational). These rules are **protocol invariants**:

1. **Sovereign primary decision:** secondary analyses (sensitivities, diagnostics, variants) never invalidate a primary decision. They can only: (i) document a limit, (ii) reveal a systematic, (iii) refine an uncertainty.
2. **Gates = veto:** each gate is blocking; a **single FAIL** $\Rightarrow$ *non-interpretable* / *no-go* result for the considered version, **regardless of** the apparent level of significance.
3. **No a posteriori promotion:** an “exploratory” metric cannot become decision-making after the fact.

Why a single primary endpoint. A single primary endpoint (here $\hat{\varepsilon}_\Phi$) deliberately reduces the space of analytical degrees of freedom: no *p-hacking*, no *garden of forking paths*, no opportunistic observable selection. This constraint is a loss of comfort, but a gain in falsifiability: *one clear question, one clear answer, and only one*.

Closure. In CHRONON, a result is published in the state in which it falls: signal, limit, or no-go. You do not “rewrite” the experiment to obtain a narrative.

2. PHILOSOPHY OF PROGRESSIVE VALIDATION

2.1. General principle

The validation of the chronon field $\Phi(x)$ relies on a logic of **controlled ascent**: a succession of independent stages, each designed as an autonomous falsification test.

No stage can be passed if the previous one has not satisfied all of its pass criteria. This discipline requires that credibility increase faster than experimental sensitivity.

The full structure comprises five levels, labeled **00 to 04**, ranging from minimal methodological credibility to a proof of network coherence.

2.2. Overview of the stages

Lvl.	Objective	Data	Observable	Dominant gates	PASS output / FAIL output
00	Methodological credibility (frozen pipeline, deterministic replication)	Public data + logs + versioned scripts	$\hat{\varepsilon}_\Phi \pm \sigma$ (declared endpoint)	Protocol integrity (preregistration, reproducibility), G6 (robustness)	PASS: complete replication pack FAIL: level 00 invalid (post-hoc / non-reproducible)
01	Controlled physical test (lever arm on X)	Vertical clock comparisons + geodesy + env.	$\hat{\varepsilon}_\Phi$ via $Y = \alpha + \varepsilon_\Phi X$	G1–G6 (all blocking)	PASS–SIGNAL or PASS–LIMIT FAIL: <i>non-interpretable / no-go</i>
02	Independent replication (coherence)	Replications (time / site / instrument / operator)	$\hat{\varepsilon}_\Phi^{(k)}$ + inter-run coherence	G2, G4, G6 (stability), inter-run coherence	PASS: coherence confirmed FAIL: inconsistency $\Rightarrow$ stop physical inference
03	Orthogonal confirmatory channel (qubits, T_2)	Series T_1, T_2, T_ϕ + controlled exposure + env.	$\hat{\beta}$ via $Y = T_2^{-1} - (T_\phi^{-1} + \frac{1}{2}T_1^{-1})$ vs X	Env. veto, null controls, permutations, DD robustness	PASS: orthogonal confirmation FAIL: systematic / published limit
04	Network signature (multi-site coherence)	Synchronized multi-site series + multiple links	Common process $s(t)$ / spectral coherence / network closure	Link independence, surrogates, band stability	PASS: coherent signature FAIL: global instrumental correlation / bound

Stop rule. If stage 01 is classified **FAIL** (at least one gate FAIL), the program is **stopped** for this version: no stages 02–04 before identifying and correcting the cause (new version, new prereg).

2.3. Stage 00 — The base camp

Objective. Establish unimpeachable metrological credibility from public data, with no new apparatus and no opportunistic tuning.

Required deliverables.

- Frozen, versioned, fully reproducible analysis pipeline.
- A single result: $\varepsilon_\Phi \pm \sigma$, accompanied by robustness plots and an explicit *gate* table.
- A complete replication pack: data, scripts, configurations, hashes.

Fundamental criterion. Any post-hoc modification invalidates the stage. Without 00, any subsequent claim amounts to opportunistic engineering.

2.4. Stage 01 — The first ascent

Objective. Constrain physics through a controlled change in local gravitational potential, maximizing the experimental lever arm on $X = g\Delta h/c^2$.

Principle. Two fixed optical clocks are compared at different heights, with an explicitly randomized measurement schedule. The only parameter of interest is the slope ε_Φ .

Admissible outcomes.

- **PASS–SIGNAL:** $\varepsilon_\Phi \neq 0$, stable under critical controls.
- **PASS–LIMIT:** $\varepsilon_\Phi \simeq 0$ with a quantitative bound.
- **SYSTEMATIC FAILURE:** a critical control fails, no physical verdict.

2.5. Stage 02 — Consolidating the route

Objective. Force alternative explanations to become specific, testable, and mutually incompatible.

Replication is performed along several independent axes:

- time (seasons, slow drifts),
- instrumental configuration,
- experimental site,
- operator.

Key criterion. Confidence intervals on ε_Φ must overlap within a tolerance defined *a priori*. A failure triggers a targeted experiment, never a rewriting of the narrative.

2.6. Stage 03 — The alternate path

Objective. Break failure modes common to clock comparisons.

The observable changes: clock frequency is replaced by quantum coherence (T_2). The exposure variable remains operationally linked to Φ .

Logical principle. An instrumental artifact specific to clocks cannot simultaneously reproduce a coherent signature in a quantum system of a different nature.

2.7. Stage 04 — The summit view

Objective. Reach a *signature*-type proof: a coherent, common, and spectrally defined pattern, observed across multiple independent nodes.

Decision criterion. A predeclared Bayesian threshold is required (e.g. $\ln B_{10} \geq 5$), with invariance with respect to links and paths.

2.8. Non-negotiable principle

In CHRONON, the **NO** must be as strong as the **YES**. This symmetry is made **operational** by a single rule:

A FAIL at any gate $\Rightarrow$ publication + stop of physical interpretation (*non-interpretable* / *no-go*) until the cause is identified.

Practical consequence. A result can be statistically “beautiful” yet scientifically null: if a gate fails, the signal is classified as *systematic* and the physical conclusion is forbidden *for this version of the protocol*. The only permitted action is: document the failure, publish, then design a targeted experiment.

What constitutes a theory change (thus forbidden here). Any post-results change that alters the tested question is considered a theory change, not a “protocol improvement”. In particular, the following are forbidden:

- changing the primary observable (Y) or the definition of exposure (X),
- changing the decision threshold (e.g. $|t| \geq 5$) or the PASS/LIMIT logic,
- changing the definition of subtracted corrections (GR, tides, geodesy) in a way driven by residuals,
- adding/removing covariates because it “helps” the result,
- reclassifying an exploratory analysis as a primary endpoint.

Only allowed exception. A **documented erratum** (bug, unit, transcription error) is allowed, provided it is traceable, published, and reproducible *without* re-tuning or optimization on the data.

3. SYSTEMATIC THREATS AND LOGICAL GATES

3.1. Role of safeguards

At extreme levels of precision, the question is not “*do we see a signal?*” but “*what could mimic this signal?*”.

CHRONON explicitly formalizes systematic threats and associates them with **logical gates**, each designed to neutralize a specific class of self-deception. A signal that does not survive a gate is considered **non-physical**.

3.2. Threat mapping

This section fixes a **threat** $\rightarrow$ **signature** $\rightarrow$ **test** $\rightarrow$ **gate** table. It is **frozen before analysis**: no threat is added/removed because “it fits better” with the results.

Threat	Expected signature if artifact	Test / data	Gate	Dominance
Slow thermal drift (clocks / links)	Apparent slope correlated with block order, time, or temperature; instability under temporal permutation	Temperature logs (multi-point), drift vs time, schedule permutations, block segmentation	G4, G5, G6	01–02
Tides / loading mis-modeled (geophysics)	Periodic residual (diurnal/semi-diurnal) or correlated with tide models; dependence on time of day	Comparison with/without <i>pre-fixed</i> corrections, spectral analysis, control on strong/weak tide windows	G4, G5, G6	01–04
Geodesy error on Δh (thus ΔU)	Biased slope varying with the measurement method, or incoherent between campaigns/site	Δh traceability, method comparison (leveling/RTK/laser), uncertainty propagation on X	G6 (robustness), (support to G1)	01–02
Clock-specific bias (A vs B)	Sign/amplitude change under swap $A \leftrightarrow B$; unstable intercept; effect attached to one clock	Full swap (high/low), identical re-analysis, compare $\hat{\varepsilon}_\Phi$ before/after swap	G2, G6	01–02
Link artifact (fiber/ GNSS/FSO/ electronics)	Slope dependence on link type or path; inconsistency across link subsets	Link-based subsets, link alternation, slope comparison by link type	G3, G6	01–04
Environmental correlation (pressure, vibrations, RF, humidity, CO ₂ if relevant)	Signal carried by a covariate (strong residual reduction when conditioning on it); strong and stable correlations	Mandatory env. logs, partial regression (diagnostic), covariate veto tests, decile analyses	G5, G6	01
Protocol correlation (measurement order, operator, maintenance)	Signal vanishes / changes under randomization; “schedule” or “operator” effect	Explicit randomization, operator/maintenance metadata, label permutations, session splits	G4, G6	01–02
Statistical overfitting / post-hoc selection	Hidden multiplicity (windows, filters, exclusions); result fragile to “reasonable” variants	Frozen-parameter journal, deterministic execution, predeclared sensitivities only	G6 + anti-post-hoc rules	00–01
Global network correlation (infrastructure, common timing)	Artificial inter-site coherence (common lines, latency, master clock); network dependence	Surrogates (time shift), alternative paths, loop closure / redundancy, line exclusions	G3 (network), surrogates (stage 04)	04

Operational reading. The *dominance* column indicates *where* the threat is most likely to mimic an

effect in X : **01** (local test), **02** (replication), **04** (network). A threat not tested by at least one dedicated gate is considered **not neutralized**.

3.3. Critical logical gates

Each gate is **blocking**: one **FAIL** $\Rightarrow$ *non-interpretable / no-go* result for the considered version. The gates use the **same definition** of (X, Y) and the **same primary model**

$$Y_i = \alpha + \varepsilon_\Phi X_i + \epsilon_i, \quad t \equiv \frac{\hat{\varepsilon}_\Phi}{\text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi)}.$$

Unless stated otherwise, gate thresholds are set to:

$$t_{\text{gate}} = 3 \quad (\text{conservative veto}), \quad k_{\text{gate}} = 2 \quad (\text{compatibility at } 2\sigma).$$

No post-hoc adjustment of these thresholds is allowed.

3.3.1. Gate G1 — Control $\Delta h \approx 0$

Minimal inputs.

- Window-by-window series: $y_{\text{mes}}(t)$, GR correction components $y_{\text{GR}}(t)$, and thus $Y(t)$.
- Exposure: $X(t)$ with $|X| \leq X_0$ (control), where X_0 is fixed *a priori* (e.g. $X_0 = 1 \times 10^{-18}$).
- Metadata: clock identifiers (A/B), link type, timestamps, quality flags.

Exact test.

- Construct the control subset $\mathcal{D}_0 = \{i : |X_i| \leq X_0\}$.
- Fit the primary model on $\mathcal{D}_0$ (same weights, same HAC).
- Windowing/filtering: identical to the primary analysis; no additional interpolation.
- Outliers: **no exclusion** based on Y ; only exclusions allowed via pre-logged quality flags.

PASS/FAIL criterion (quantified).

$$\text{PASS} \iff |t_0| < t_{\text{gate}} \quad \text{where} \quad t_0 = \frac{\hat{\varepsilon}_{\Phi,0}}{\text{SE}_{\text{HAC}}(\hat{\varepsilon}_{\Phi,0})}.$$

$$\text{FAIL} \iff |t_0| \geq t_{\text{gate}}.$$

Expected evidence (required artifacts).

- **Figure G1**: scatter Y vs X on $\mathcal{D}_0$ + fit + CI band.
- **Table G1**: $\hat{\varepsilon}_{\Phi,0}$, SE_{HAC} , t_0 , N_0 , PASS/FAIL status.

3.3.2. Gate G2 — Permutation $A \leftrightarrow B$

Minimal inputs.

- Two full executions of the pipeline: (i) nominal convention, (ii) swap $A \leftrightarrow B$ (high/low inverted).
- Series (X_i, Y_i) and estimators $\hat{\varepsilon}_\Phi$, $\hat{\varepsilon}_\Phi^{\text{swap}}$ with their HAC SEs.

Exact test.

- Run the pipeline *identically* after the swap (same windows, same weights, same corrections).
- Compute

$$\Delta \hat{\varepsilon}_\Phi \equiv \hat{\varepsilon}_\Phi^{\text{swap}} - \hat{\varepsilon}_\Phi, \quad \text{SE}(\Delta) = \sqrt{\text{SE}_{\text{HAC}}^2(\hat{\varepsilon}_\Phi^{\text{swap}}) + \text{SE}_{\text{HAC}}^2(\hat{\varepsilon}_\Phi)}.$$

- Outliers: same rules (quality flags only), no manual tweaking.

PASS/FAIL criterion (quantified).

$$\text{PASS} \iff |\Delta \hat{\varepsilon}_\Phi| \leq k_{\text{gate}} \text{SE}(\Delta) \quad \text{and} \quad \left(\text{sign}(\hat{\varepsilon}_\Phi) = \text{sign}(\hat{\varepsilon}_\Phi^{\text{swap}}) \text{ or } \min(|t|, |t_{\text{swap}}|) < 1 \right).$$

$$\text{FAIL} \iff \text{non-PASS}.$$

Expected evidence (required artifacts).

- **Figure G2:** Y vs X with two overlaid fits (nominal vs swap) + CI.
- **Table G2:** $\hat{\varepsilon}_\Phi$, SE_{HAC} , t (nominal); same (swap); $\Delta\hat{\varepsilon}_\Phi$, $\text{SE}(\Delta)$; PASS/FAIL.

*3.3.3. Gate G3 — Link permutation / independence***Minimal inputs.**

- Series annotated by link type (fiber/GNSS/FSO/...) and/or path.
- Subsets $\mathcal{D}_\ell$ per link ℓ with sufficient N_ℓ (threshold fixed *a priori*, e.g. $N_\ell \geq 8$ windows).

Exact test.

- Estimate $\hat{\varepsilon}_\Phi^{(\ell)}$ and $\text{SE}_{\text{HAC}}^{(\ell)}$ on each $\mathcal{D}_\ell$ (same pipeline).
- Test heterogeneity with a Cochran-type statistic:

$$Q = \sum_{\ell} w_{\ell} \left(\hat{\varepsilon}_\Phi^{(\ell)} - \hat{\varepsilon}_\Phi^{\text{all}} \right)^2, \quad w_{\ell} = \frac{1}{\left(\text{SE}_{\text{HAC}}^{(\ell)} \right)^2}.$$

- Outliers: no ad hoc exclusion; same quality flags.

PASS/FAIL criterion (quantified).

$$\text{PASS} \iff p(Q) \geq 0.01 \quad \text{and} \quad \max_{\ell, \ell'} \left| \hat{\varepsilon}_\Phi^{(\ell)} - \hat{\varepsilon}_\Phi^{(\ell')} \right| \leq k_{\text{gate}} \sqrt{(\text{SE}^{(\ell)})^2 + (\text{SE}^{(\ell')})^2}.$$

$$\text{FAIL} \iff \text{non-PASS}.$$

Expected evidence (required artifacts).

- **Figure G3:** forest plot of $\hat{\varepsilon}_\Phi^{(\ell)}$ (CI) + global estimate.
- **Table G3:** per link: N_ℓ , $\hat{\varepsilon}_\Phi^{(\ell)}$, SE , t_ℓ ; then Q , $p(Q)$; PASS/FAIL.

*3.3.4. Gate G4 — Schedule randomization***Minimal inputs.**

- Actual schedule (timestamps, block order, ascents/descents) + randomization seed (if applicable).
- Series (X_i, Y_i) per block, and session metadata.

Exact test.

- Construct M schedule permutations (at minimum $M = 1000$), permuting the temporal order of blocks while preserving block structure (same durations).
- For each permutation m , recompute the estimator $\hat{\varepsilon}_\Phi^{(m)}$ (same pipeline, same parameters).
- Quantify schedule sensitivity:

$$\sigma_{\text{perm}} \equiv \text{SD}(\{\hat{\varepsilon}_\Phi^{(m)}\}_{m=1}^M), \quad \Delta_{\text{perm}} \equiv \left| \hat{\varepsilon}_\Phi - \text{median}(\hat{\varepsilon}_\Phi^{(m)}) \right|.$$

- Outliers: identical rules; no targeted removal.

PASS/FAIL criterion (quantified).

$$\text{PASS} \iff \Delta_{\text{perm}} \leq k_{\text{gate}} \sigma_{\text{perm}} \quad \text{and} \quad \sigma_{\text{perm}} \leq \text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi).$$

$$\text{FAIL} \iff \text{non-PASS}.$$

Expected evidence (required artifacts).

- **Figure G4:** histogram of $\hat{\varepsilon}_\Phi^{(m)}$ (permutations) + vertical line at observed $\hat{\varepsilon}_\Phi$.
- **Table G4:** M , median, σ_{perm} , Δ_{perm} , $\text{SE_HAC}(\text{observed})$, PASS/FAIL.

3.3.5. Gate G5 — Veto by environmental covariates

Minimal inputs.

- Synchronized environmental logs (**fixed** list: temperature (multi-point), pressure, humidity, vibrations, RF/EMI, power/UPS, etc.).
- Series (X_i, Y_i) temporally aligned with covariates $Z_{j,i}$.

Exact test.

- Check potential confounding: correlation between X and each covariate Z_j .

$$\rho_j = \text{corr}(X, Z_j).$$

- Fit the extended (diagnostic) model:

$$Y_i = \alpha + \varepsilon_\Phi X_i + \sum_{j=1}^p \gamma_j Z_{j,i} + \epsilon_i$$

while keeping the same windowing and the same HAC policy.

- Outliers: no exclusion based on Y ; only exclusions via quality flags (explicitly logged sensor failures).

PASS/FAIL criterion (quantified).

$$\text{PASS} \iff \max_j |\rho_j| \leq 0.30 \quad \text{and} \quad \left| \hat{\varepsilon}_\Phi^{(Z)} - \hat{\varepsilon}_\Phi \right| \leq k_{\text{gate}} \sqrt{\text{SE}^2(\hat{\varepsilon}_\Phi^{(Z)}) + \text{SE}^2(\hat{\varepsilon}_\Phi)}.$$

$$\text{FAIL} \iff \text{non-PASS}.$$

(Here $\hat{\varepsilon}_\Phi^{(Z)}$ denotes the estimate in the covariate model.)

Expected evidence (required artifacts).

- **Figure G5:** (i) correlation heatmap ρ_j ; (ii) comparison $\hat{\varepsilon}_\Phi$ vs $\hat{\varepsilon}_\Phi^{(Z)}$ (CI bars).
- **Table G5:** list of Z_j (unit), ρ_j , $\hat{\varepsilon}_\Phi^{(Z)}$, SE, Δ , PASS/FAIL.

3.3.6. Gate G6 — Statistical robustness

Minimal inputs.

- Full dataset (X_i, Y_i) , weights/uncertainties, quality flags, and primary-model output.
- **Predeclared** robustness parameters (number of bootstrap draws, etc.).

Exact test. We run a suite of *non-adaptive* robustness tests:

- **Leave-one-out (LOO):** recompute $\hat{\varepsilon}_\Phi^{(-i)}$ for each point/block i .
- **Wild bootstrap** (diagnostic): B draws (e.g. $B = 2000$) to obtain a dispersion σ_{boot} .
- **Robust regression** (Huber) as a diagnostic *without* removing points.

Windowing/filtering identical to the primary. Outliers: **no removal** driven by Y .

PASS/FAIL criterion (quantified). Define:

$$p_{\text{flip}} \equiv \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{sign}(\hat{\varepsilon}_\Phi^{(-i)}) \neq \text{sign}(\hat{\varepsilon}_\Phi)], \quad \Delta_{\text{LOO}} \equiv \max_i \left| \hat{\varepsilon}_\Phi^{(-i)} - \hat{\varepsilon}_\Phi \right|.$$

$$\text{PASS} \iff p_{\text{flip}} \leq 0.05 \quad \text{and} \quad \Delta_{\text{LOO}} \leq k_{\text{gate}} \text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi) \quad \text{and} \quad \sigma_{\text{boot}} \leq 1.5 \text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi).$$

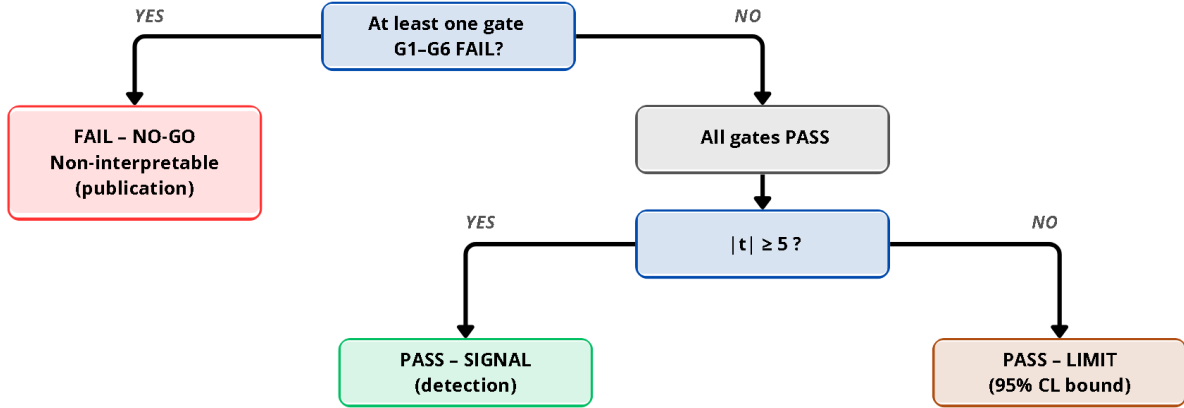
$$\text{FAIL} \iff \text{non-PASS}.$$

Expected evidence (required artifacts).

- **Figure G6:** (i) distribution of $\hat{\varepsilon}_\Phi^{(-i)}$ (LOO); (ii) bootstrap distribution (if executed).
- **Table G6:** p_{flip} , Δ_{LOO} , σ_{boot} , robust vs primary comparison, PASS/FAIL.

3.4. Global decision logic

The decision follows a strict logical chain. It does not rely on a single metric, but on a frozen **decision tree**:



Definition of decision metrics.

$$t \equiv \frac{\hat{\epsilon}_{\Phi}}{\text{SE}_{\text{HAC}}(\hat{\epsilon}_{\Phi})}.$$

Gates G1–G6 have their own criteria (Section 3.3) and are **veto**es.

Frequentist vs Bayesian conflict rule (if both are reported). CHRONON allows simultaneous reporting of a frequentist test (t) and a Bayesian score (e.g. $\ln B_{10}$), but **the decision is conservative** and follows the rule below:

1. **If any gate FAIL:** verdict = **FAIL–NO-GO** (regardless of t or $\ln B_{10}$).
2. **If all gates PASS:**
 - **PASS–SIGNAL** only if *all* decision criteria are met (e.g. $|t| \geq 5$ and $\ln B_{10} \geq 5$ if Bayesian is declared decision-making).
 - Otherwise, verdict = **PASS–LIMIT**.

Principle. In case of disagreement, **we never promote** a result to a “signal”. We publish a **limit** (or a **no-go** if a gate FAILs), and the frequentist/Bayesian gap is documented as a diagnostic, not as a discovery argument.

4. STAGE 01 — CONTROLLED PHYSICAL TEST USING OPTICAL CLOCKS

4.1. Scientific objective

Stage 01 is the first direct confrontation with physical reality. The objective is to test the existence of a non-metric residual after complete subtraction of the General Relativity prediction, by maximizing the experimental lever arm on gravitational exposure

$$X \equiv \frac{g \Delta h}{c^2}.$$

This stage does not seek a generalization, but a local, controlled, and falsifiable test.

4.2. Experimental principle

Two fixed optical clocks (A and B) are compared at distinct heights in a quasi-stationary terrestrial gravitational field. The GR prediction

$$y_{\text{GR}} = \frac{g \Delta h}{c^2}$$

is computed from traceable geodesy, then subtracted point-by-point.

The only analyzed observable is then:

$$Y = y_{\text{mes}} - y_{\text{GR}}.$$

Any residual linear dependence of Y on X is either a systematic, or a signal of new physics.

4.3. Recommended geometry

Vertical configuration.

- Stairwell, tower, or rigid vertical structure.
- Typical span: $\Delta h \in [10, 30]$ m.
- Multiple levels (at least 3–5) to improve slope identifiability.

Justification.

- The gravitational field is quasi one-dimensional.
- Relativistic corrections are well known and stable.
- Environmental gradients are measurable and controllable.

4.4. Instrumentation

Clocks.

- Stabilized optical clocks (Sr, Yb, or equivalent).
- Continuous operation over windows ≥ 1 h.

Frequency link.

- Phase-stabilized optical fiber (reference).
- Alternative links (GNSS, free-space) used for permutations.

Geodesy.

- Measurement of Δh by optical leveling, RTK, or laser.
- Explicit corrections: solid Earth tides, ocean tides, atmospheric loading.

4.5. Measurement protocol

The measurement protocol is frozen before acquisition. No adjustment (durations, order, exclusions, sensors) is allowed after the start, except for a **logged incident** and a **documented erratum**.

1. **Prior definition of height levels.** Levels $\{\Delta h_j\}$ (at least 3–5) are defined and measured before the first comparison, with documented uncertainties.
2. **Full randomization of measurement order.** The order of blocks (levels) is randomized; the randomization seed is archived.
3. **Alternating ascents / descents (operational swap).** The “high” / “low” roles are inverted according to a predeclared schedule, to feed G2.
4. **Control blocks $\Delta h \approx 0$ (formalized).** Controls are mandatory and integrated *into* the schedule:
 - Minimum frequency: at least $f_0 = 20\%$ of blocks (in acquisition time) are controls.
 - Minimum duration: each control lasts at least $T_0 = 30$ minutes (or ≥ 1 analysis window).
 - Placement: interleaved and randomized controls (not “only at the beginning” nor “only at the end”).
 - Operational definition: $|X| \leq X_0$ (threshold fixed a priori, e.g. $X_0 = 1 \times 10^{-18}$).
5. **Environmental logging (mandatory, fixed list).** The following variables are recorded continuously (common timestamp):
 - Temperature: at least 3 points (clock source, electronics, link path),
 - Pressure, humidity,
 - Vibrations (accelerometer) or documented proxy,
 - RF/EMI noise (proxy: broadband level, or instrumentation state indicator),
 - Power / UPS status (voltage, events),
 - System states: locks, loss-of-lock, maintenance, resets,
 - Link metadata: type, path, events (cuts, switchovers).

Data quality / exclusions (frozen rules). Exclusions are **allowed only** if they satisfy the two conditions below:

1. **Exclusion based on a non- Y criterion:** loss of lock, instrument flag, sensor failure, inconsistent timestamp, or logged technical event.
2. **Traceability:** each exclusion is **logged**, justified, timestamped, and reproducible (same rule $\Rightarrow$ same exclusion).

Explicitly **forbidden** are:

- exclusions “because the point is far from the fit”,
- window cutting chosen after observing $\hat{\varepsilon}_\Phi$,
- removal of an entire block with no corresponding technical event.

Protocol output. The final executed schedule, the randomization seed, and the full log (env. + instrument + link) are archived as evidence of protocol compliance.

4.6. Primary analysis

The primary analysis is single, frozen, and executable identically. It produces the estimator $\hat{\varepsilon}_\Phi$ from a residual Y defined by **subtraction** of predeclared corrections.

Exact definition of the residual. For each window (or block) indexed by i :

$$Y_i \equiv y_{\text{mes},i} - y_{\text{GR},i}, \quad X_i \equiv \frac{\Delta U_i}{c^2} \simeq \frac{g \Delta h_i}{c^2}.$$

The quantity $y_{\text{GR},i}$ includes the relativistic gravitational correction (redshift) computed from ΔU_i and *only* the GR terms predeclared in this document. Any additional non-predeclared correction is forbidden (except documented erratum).

Windows, time alignment, interpolation (frozen rules).

- **Analysis window:** data are aggregated into windows of duration T_{win} fixed *a priori* (e.g. $T_{\text{win}} = 1800\text{s}$), not optimized on the data.
- **Alignment:** all series (clocks, link, geodesy, environment) are aligned on a common time grid (UTC) by timestamp.
- **Interpolation: forbidden** for Y and X (no “gap filling”). Only mean/median aggregation over the window is allowed if coverage exceeds a threshold η_{min} fixed *a priori* (e.g. $\eta_{\text{min}} = 90\%$ of points present).
- **Outliers:** no removal based on the value of Y . The only allowed exclusions follow the rules of Section 4.5 (non- Y quality flags).

Primary model (single). The primary fit is:

$$Y_i = \alpha + \varepsilon_\Phi X_i + \epsilon_i,$$

with free intercept α . The slope ε_Φ is the single parameter of interest. Weights and error estimation (HAC) are detailed in Section 5.3.

Units & normalizations.

- X is **dimensionless**: $X = \Delta U/c^2$ (or $g\Delta h/c^2$ in the local approximation).
- y is a **fractional frequency**: $y = \delta\nu/\nu$ (dimensionless), hence Y is also dimensionless.
- ε_Φ is **dimensionless** (residual slope).

Primary output. The primary analysis must produce at minimum:

$$\hat{\varepsilon}_\Phi, \quad \text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi), \quad t = \hat{\varepsilon}_\Phi/\text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi), \quad \text{and the PASS/FAIL status of the gates.}$$

4.7. Decision logic for stage 01

Stage 01 admits exactly three outcomes. They are **mutually exclusive** and **frozen** by the rules below.

1) PASS–SIGNAL (admissible detection). The **PASS–SIGNAL** outcome is declared if, and only if:

1. **All** critical gates G1–G6 are **PASS** (Section 3.3), and
2. the primary frequentist threshold is crossed:

$$|t| \geq 5, \quad t \equiv \frac{\hat{\varepsilon}_\Phi}{\text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi)}.$$

In this case, the result is published as a *local detection* (stage 01), accompanied by the full audit pack (figures/tables/logs/checksums), and serves as input to stage 02 (replication).

2) PASS–LIMIT (quantitative constraint). The **PASS–LIMIT** outcome is declared if, and only if:

1. **All** critical gates G1–G6 are **PASS**, and
2. the detection threshold is not crossed:

$$|t| < 5.$$

In this case, a 95% CL bound is published:

$$|\varepsilon_\Phi| < 1.96 \text{ SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi),$$

and stage 02 is undertaken only if it was predeclared as necessary to improve the bound (otherwise the “limit” outcome closes stage 01).

3) FAIL / NON-INTERPRETABLE (operational no-go). The **FAIL–NO-GO** outcome is declared if:

$$\exists \text{ gate } G_j \in \{G1, \dots, G6\} \text{ such that } G_j = \mathbf{FAIL}.$$

Consequences:

- any estimate of $\hat{\varepsilon}_\Phi$ is classified as *non-interpretable* for a physical conclusion,
- the result is **published** as a systematic failure (with diagnostic),
- the program is **stopped** for this version (Stop rule) until the cause is identified and a new preregistered version is defined.

Non-rescue rule. No secondary analysis, variant, or alternative metric can turn a **FAIL–NO-GO** into **PASS** (signal or limit).

5. STATISTICAL FRAMEWORK, SENSITIVITY, AND DECISION RULES

5.1. Guiding principle

CHRONON adopts a simple and non-negotiable rule:

A single primary endpoint, a single primary model, and a pre-defined binary decision.

Anything that falls outside this framework is classified as *diagnostic* or *sensitivity*, and has no confirmatory power.

5.2. Primary statistical model

For each measurement window i :

$$Y_i = \alpha + \varepsilon_\Phi X_i + \epsilon_i, \quad X_i = \frac{g \Delta h_i}{c^2}.$$

- Free intercept α (stability policy).
- Weighting by $1/\sigma_{Y,i}^2$, if available, otherwise OLS.
- Heteroskedastic and time-correlated errors.

5.3. Estimation and errors

This section locks the notation and the estimation procedure. No post-hoc adjustment (HAC choice, lags, kernel, exclusions) is allowed.

Notation (standardized).

- Physical parameter (target): ε_Φ .
- Estimator (data): $\hat{\varepsilon}_\Phi$.
- Primary standard error: $\text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi)$.
- Test statistic: $t \equiv \hat{\varepsilon}_\Phi / \text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi)$.

Main estimator.

- **WLS** regression if uncertainties $\sigma_{Y,i}$ are available: weights $w_i = 1/\sigma_{Y,i}^2$.
- Otherwise, **OLS** (uniform weights) with the same HAC error policy.
- Free intercept α .

HAC (heteroskedasticity + autocorrelation): frozen definition. We use a Newey–West HAC estimator with:

- **Kernel:** Bartlett (triangular), fixed.
- **Lag/bandwidth:** Andrews automatic rule (data-driven *but* fixed as a rule, not modifiable after the fact).

The selected lag is explicitly reported as L_{NW} in the outputs. Any change of kernel or lag rule is a **protocol change** (forbidden in this version).

Mandatory sensitivities (non-decision-making). These analyses are executed and reported *as diagnostics*:

- Newey–West lag divided by 2, and multiplied by 2: $L_{\text{NW}}/2$ and $2L_{\text{NW}}$.
- Wild bootstrap (diagnostic) with B draws fixed *a priori* (e.g. $B = 2000$) for a dispersion σ_{boot} .
- Robust regression (Huber) *without* removing points.

None of these sensitivities can promote a primary decision.

Uncertainty on X (if non-negligible). If $\sigma_X/X_{\text{rms}} \geq 0.1$ (fixed threshold), a Deming regression (WTLS) is run as a control, with error ratio $\lambda = \sigma_Y^2/\sigma_X^2$ fixed *a priori*. The decision verdict remains that of the primary WLS/OLS + HAC model, unless Deming reveals an instability classified under G6 (robustness) as FAIL.

Required output. Each run must report:

$\hat{\varepsilon}_\Phi$, $\text{SE}_{\text{HAC}}(\hat{\varepsilon}_\Phi)$, t , L_{NW} , and the sensitivities (lag/2, lag $\times$ 2, bootstrap, Huber) labeled.

5.4. Expected sensitivity

To first order, the uncertainty on the slope is:

$$\sigma_{\hat{\varepsilon}_\Phi} \approx \frac{\sigma_Y}{\sqrt{N}} \frac{c^2}{g \Delta h_{\text{rms}}}.$$

Typical orders of magnitude:

- $\Delta h_{\text{rms}} \sim 10\text{--}30$ m,
- $N \sim 20\text{--}60$ permutations,
- $\sigma_Y \sim 10^{-18}$ to 10^{-19} .

This corresponds to a sensitivity on $|\varepsilon_\Phi|$ between 10^{-3} and 10^{-5} , depending on metrological quality.

5.5. Intercept policy

- α must be stable under A/B permutations, sites, and operators.
- $\alpha = 0 \pm \sigma$ is required on the controls $\Delta h \approx 0$.
- A nonzero $\alpha \neq 0$ outside controls is tolerated if it is stable and interpretable as an instrumental offset. An unstable intercept invalidates any physical interpretation.

5.6. Decision threshold

The decision threshold is **preregistered** and **not optimized** on the data. It is chosen conservatively to minimize false detections in a context dominated by systematic-risk.

Frequentist (primary endpoint). The primary statistic is:

$$t = \frac{\hat{\epsilon}_{\Phi}}{\text{SE}_{\text{HAC}}(\hat{\epsilon}_{\Phi})}.$$

Rule:

- **Signal candidate** if $|t| \geq 5$,
- **Otherwise:** limit (if gates PASS) or no-go (if any gate FAIL).

Bayesian (if declared decision-making). A Bayes factor $\ln B_{10}$ is also reported. The threshold

$$\ln B_{10} \geq 5$$

is a conservative preregistered threshold, *not optimized on the data*, corresponding to a requirement of strong evidence before any signal claim.

Combined rule (clear and conservative). If $\ln B_{10}$ is used as a decision metric (and not only descriptive), then the **PASS–SIGNAL** condition becomes:

$$\text{PASS–SIGNAL} \iff \left(\forall j \in \{1, \dots, 6\}, G_j = \text{PASS} \right) \wedge (|t| \geq 5) \wedge (\ln B_{10} \geq 5).$$

In all other cases with gates PASS, the outcome is **PASS–LIMIT**. With at least one gate FAIL, the outcome is **FAIL–NO-GO**.

Conflict rule. In case of conflict ($|t| \geq 5$ but $\ln B_{10} < 5$, or the reverse), the result is **not** promoted to a signal: a **limit** is published and the discrepancy is documented as a diagnostic.

5.7. Mandatory outputs

Each pipeline run produces an **executable** replication pack. The absence of a single element makes the result **non-compliant** (thus non-decision-making).

1) Data and formats (export).

- `data_analysis.csv` (or `.parquet`): final window-by-window table containing at minimum

$$\{t_i, X_i, Y_i, \sigma_{Y,i} \text{ (if available), clocks A/B, link, } \Delta h, \Delta U, \text{ quality flags}\}.$$

- `env_log.csv`: mandatory environmental variables (Section 4.5) with timestamp.
- `geo_log.csv`: geodesy (Δh , method, uncertainties, applied corrections).

2) Scripts, versioning, reproducibility.

- `run.sh` (or equivalent): single command to reproduce all outputs.
- Source code + **git version** (commit hash): `GIT_COMMIT.txt`.
- Frozen configuration: `config.yaml` (windows, thresholds, HAC kernel, Andrews rule, X_0 , f_0 , etc.).

3) Checksums (integrity proof).

- `SHA256SUMS.txt` listing SHA256 hashes of: raw data (if publishable), analysis data, scripts, config, figures, tables.

4) Mandatory tables (results and audit).

- **Primary result table:** $\hat{\epsilon}_\Phi$, $\text{SE}_{\text{HAC}}(\hat{\epsilon}_\Phi)$, t , 95% CI, L_{NW} .
- **Gates table:** for each G1–G6: statistic(s), quantified criterion, PASS/FAIL, reference to associated figure/table.
- **Frozen vs measured parameters table:**
 - *Frozen:* T_{win} , η_{min} , t^* , t_{gate} , k_{gate} , X_0 , f_0 , T_0 , HAC kernel, Andrews rule, B , M , etc.
 - *Measured:* g , Δh , geodetic uncertainties, loss-of-lock rates, sensor coverage, etc.

5) Mandatory figures.

- **Primary figure:** Y vs X (free-intercept fit) + uncertainty band + annotation of $\hat{\epsilon}_\Phi$.
- **Residual diagnostics:** residuals vs time, residuals vs prediction, and histogram (or QQ-plot) of residuals.
- **Gate figures:** at least 1 figure per gate (G1–G6), as specified in Section 3.3.

6) Exclusion log (if applicable).

- **exclusions.csv:** each exclusion with timestamp, triggered rule, evidence (flag/event), and source identifier (instrument/link/sensor).

Status. This pack constitutes the *final scientific verdict* of the considered stage: without it, no claim (signal, limit, or no-go) is admissible.

6. STAGE 02 — REPLICATION AND CONSOLIDATION

6.1. Rationale

A local signal, even clean and statistically significant, remains scientifically fragile as long as it has not survived independent conditions.

Stage 02 does not seek to improve significance, but to **destroy accidental explanations**. It turns a local result into a structured constraint.

6.2. Replication axes

Replication is conducted along orthogonal axes, each targeting a distinct class of bias.

Axis	What it eliminates
Time	slow drifts, seasonal correlations
Site	local specificities (gravity, environment)
Instrument	clock-specific defects
Operator	human biases and implicit procedures
Configuration	geometric or software dependencies

No axis is optional. A single unexplored axis limits the global interpretation.

6.3. Replication formalism

Each replication k provides an estimate:

$$\hat{\epsilon}_\Phi^{(k)} \pm \sigma_k.$$

Estimates are compared by:

- their sign,
- their confidence-interval overlap,
- their stability under the primary gates.

A coherence failure is not masked by a global average.

6.4. Minimal coherence criterion

Inter-replication coherence is **quantified** and **blocking**. It cannot be “rescued” by a global average.

Each replication k provides:

$$\hat{\varepsilon}_{\Phi}^{(k)} \pm \text{SE}^{(k)} \quad \text{with} \quad \text{SE}^{(k)} = \text{SE}_{\text{HAC}}\left(\hat{\varepsilon}_{\Phi}^{(k)}\right),$$

and a gates status (at minimum G1–G6) which must be **PASS** to be admissible.

Quantified tolerance (pairwise compatibility). We impose pairwise compatibility:

$$\forall(k, \ell), \quad \left| \hat{\varepsilon}_{\Phi}^{(k)} - \hat{\varepsilon}_{\Phi}^{(\ell)} \right| \leq k_{\text{rep}} \sqrt{\left(\text{SE}^{(k)}\right)^2 + \left(\text{SE}^{(\ell)}\right)^2},$$

with a threshold fixed *a priori*: $k_{\text{rep}} = 2$ (compatibility at 2σ).

Sign coherence (required). If at least two replications are informative in the sense

$$|t^{(k)}| \geq 2 \quad \text{with} \quad t^{(k)} = \hat{\varepsilon}_{\Phi}^{(k)} / \text{SE}^{(k)},$$

then the sign must be invariant:

$$\text{sign}\left(\hat{\varepsilon}_{\Phi}^{(k)}\right) = \text{sign}\left(\hat{\varepsilon}_{\Phi}^{(\ell)}\right) \quad \text{for all informative pairs } (k, \ell).$$

If all replications are non-informative ($|t^{(k)}| < 2$ for all), sign coherence is not required (“limit” case), but amplitude compatibility remains mandatory.

Coherence decision.

- **COHERENCE VALIDATED** if all pairs satisfy the tolerance and, if applicable, sign coherence.
- **INCOHERENCE** otherwise $\Rightarrow$ stop global physical inference (publication + diagnostic).

Mandatory outputs.

- **Figure:** forest plot of $\hat{\varepsilon}_{\Phi}^{(k)}$ with CI (and gates status).
- **Table:** matrix of normalized pairwise gaps $D_{k\ell} = \frac{\left| \hat{\varepsilon}_{\Phi}^{(k)} - \hat{\varepsilon}_{\Phi}^{(\ell)} \right|}{\sqrt{(\text{SE}^{(k)})^2 + (\text{SE}^{(\ell)})^2}} + \text{coherence verdict}$.

6.5. Aggregation (if admissible)

Aggregation is **not a right**. It is an operation allowed **only** if minimal coherence (Section 6.4) is validated.

Aggregation condition (hard-coded). We aggregate **only** if:

1. all aggregated replications have **all gates PASS**, and
2. minimal pairwise coherence is satisfied (Section 6.4), and
3. overall heterogeneity is low in the sense:

$$p(Q) \geq 0.05, \quad Q = \sum_k w_k \left(\hat{\varepsilon}_{\Phi}^{(k)} - \hat{\varepsilon}_{\Phi}^{\text{FE}} \right)^2, \quad w_k = \frac{1}{\left(\text{SE}^{(k)}\right)^2}.$$

If a single condition fails, aggregation is **forbidden**: results are published separately (and the “heterogeneity/incoherence” verdict).

Aggregation model (frozen). CHRONON uses a **fixed-effects** model:

$$\hat{\varepsilon}_{\Phi}^{\text{FE}} = \frac{\sum_k w_k \hat{\varepsilon}_{\Phi}^{(k)}}{\sum_k w_k}, \quad \text{SE}\left(\hat{\varepsilon}_{\Phi}^{\text{FE}}\right) = \left(\sum_k w_k \right)^{-1/2}.$$

This choice is frozen because it *rewards* coherence only: if the effect varies, we do not aggregate.

Anti-magic-average rule. No random-effects model is used to “save” a heterogeneous set. Heterogeneity is a result in itself: it triggers diagnostics and a stop of global inference, not a rhetorically weakened average.

Mandatory outputs.

- **Aggregation table:** $\hat{\varepsilon}_{\Phi}^{(k)}$, $\text{SE}^{(k)}$, w_k , relative contribution, Q , $p(Q)$, status “aggregation allowed / forbidden”.
- **Figure:** forest plot + line $\hat{\varepsilon}_{\Phi}^{\text{FE}}$ (if allowed).

6.6. Possible outcomes of stage 02

Stage 02 is meant to decide the following question: *is the stage 01 result universal (replicable) or contingent (accidental)?*

Outcomes are strict and **anti-magic-average**.

Non-negotiable rule (publication + stop).

An incoherent replication (Section 6.4) $\Rightarrow$ **publication + stop of global physical inference:** *no average, no “global signal”, no statistical rescue.*

1) COHERENCE CONFIRMED (passage allowed). This outcome is declared if, and only if:

1. all considered replications have **all gates PASS**, and
2. quantified minimal coherence (Section 6.4) is validated, and
3. if an aggregation is reported: aggregation is **allowed** (Section 6.5).

Consequence: passage allowed to stage 03 (orthogonal confirmatory channel). Passage to stage 04 is allowed only after at least one orthogonal validation (stage 03) or a preregistered justification specific to the network.

2) DIAGNOSTIC HETEROGENEITY (targeted program). This outcome is declared if:

- gates PASS are satisfied *but*
- minimal coherence fails **without** isolating a single unique artifact.

Consequence: publication of separate results + launch of targeted experiments to test causes (site/instrument/season). No stage 03/04 is undertaken until the origin of heterogeneity is identified.

3) INCOHERENCE (abandon global physical interpretation). This outcome is declared if:

- at least one replication is incompatible in the sense of Section 6.4,
- or if the sign is not stable when required,
- or if a replication fails a critical gate.

Consequence: the stage 01 signal is classified as *non-universal / probable systematic*. The incoherence is published and we **do not extrapolate** toward a fundamental tempo field.

Exact conditions for moving to 03 and 04.

- **Move to 03:** only if the outcome is **COHERENCE CONFIRMED**.
- **Move to 04:** only if (i) **COHERENCE CONFIRMED** and (ii) stage 03 provides at minimum a compatible limit or an orthogonal confirmation, under the conservative conflict rule (no “network” to rescue an incoherence).

6.7. Scientific value of a failure

A failure at stage 02 is informative. It allows one to:

- map regimes of instrumental validity,
- identify local non-universal effects,
- publish a robust negative constraint.

In CHRONON, replication is not used to convince. It is used to eliminate.

7. STAGE 03 — ORTHOGONAL OBSERVABLE: QUANTUM COHERENCE

7.1. Physical motivation

Optical clock comparisons, however rigorous they may be, share a common structure: classical oscillators stabilized in frequency, metrological links, and explicit relativistic corrections.

Stage 03 introduces a **change in the nature of the observable**. Frequency is abandoned in favor of *quantum coherence*, to break any class of biases specific to clocks.

If an effect is real and linked to the local tempo $\Phi(x)$, it must manifest in a quantum system through a modification of its decoherence laws.

7.2. Target system

Admissible platforms include:

- superconducting qubits (cQED),
- trapped ions,
- solid-state spins (NV, equivalents).

The criterion is not the technology, but the joint availability of:

- coherence time T_2 ,
- relaxation T_1 and pure dephasing T_ϕ ,
- a controlled displacement or exposure protocol.

7.3. Operational model

The adopted coherence-loss law is:

$$T_2^{-1} = T_\phi^{-1} + \frac{1}{2}T_1^{-1} + \beta \Delta \ln \Phi, \quad \Delta \ln \Phi \simeq \frac{g \Delta h}{c^2}.$$

- β is a signed coefficient.
- $\Delta \ln \Phi$ is dimensionless and operational.

This writing guarantees:

- no modification of Hamiltonian dynamics,
- no additional energy term,
- a strictly differential interpretation.

7.4. Admissible protocols

Three protocol classes are accepted:

Static vertical stacking. Qubits placed at different fixed heights, with prolonged exposure and stabilized environment.

Controlled displacement. Elevator, winch, or mobile platform, with randomized ascent/descent alternation.

Acceleration profile. Effective exposures via controlled acceleration (parabola, centrifuge), with reconstruction of $\Delta \ln \Phi_{\text{eff}}$.

7.5. Statistical analysis

This stage fixes **the metric, the model, and the link to X** . No variant (lin/log, other metric) can become decision-making after the fact.

Primary metric (frozen). We work on the **excess decoherence rate**:

$$Y_k \equiv T_{2,k}^{-1} - \left(T_{\phi,k}^{-1} + \frac{1}{2}T_{1,k}^{-1} \right),$$

where k indexes blocks/measurements. This definition removes (to first order) the expected contributions of T_1 and pure dephasing, and isolates a testable residual.

Link to exposure (same variable X , same convention). We impose the same operational exposure as for clocks:

$$X_k \equiv \frac{\Delta U_k}{c^2} \simeq \frac{g \Delta h_k}{c^2},$$

with the same sign conventions ($\Delta h > 0 \Rightarrow X > 0$). Time shifting between Y_k and X_k is **forbidden**: each block k is associated with the *block-mean* exposure (same timestamps). Any “a posteriori” reconstruction of a more favorable X_k is forbidden.

Primary statistical model (linear). The primary fit is:

$$Y_k = \alpha + \beta X_k + \epsilon_k,$$

with free intercept α . The slope β has units of s^{-1} . If a log dependence is desired, it constitutes a **different protocol** (forbidden in this version).

Windows, filtering, exclusions (frozen rules).

- Windows: blocks of duration $T_{\text{win}}^{(q)}$ fixed *a priori* (e.g. 10–30 min), identical for all exposure levels.
- Filtering: no adaptive filtering; only planned standard sequences (Ramsey/echo/DD).
- Exclusions: only on non- Y criteria (loss of lock, sequence failure, dead sensor), logged and reproducible; never “because T_2 is weird”.

Internal null tests (blocking). Two placebo tests are mandatory:

1. **Sequences without gradient (placebo $X \simeq 0$).** We run blocks where exposure does not vary ($|X_k| \leq X_0^{(q)}$ fixed), and require:

$$\hat{\beta}_0 = 0 \pm \sigma, \quad |t_0| < t_{\text{gate}}.$$

2. **Label inversion / permutation.** We randomly permute the labels X_k (or the order of blocks) over M permutations (e.g. $M = 1000$) to verify that the result is not carried by a hidden temporal correlation.

Failure of a null test classifies stage 03 as **systematic** (non-interpretable).

Mandatory outputs (audit).

- Primary figure: Y_k vs X_k + fit + CI.
- Placebo figure: distribution of $\hat{\beta}^{(m)}$ under permutations + position of observed $\hat{\beta}$.
- Table: $\hat{\beta}$, SE (HAC if time correlation), t , null-test status, and execution logs.

7.6. Critical controls

The following controls are **blocking**:

- **DD robustness**: the slope β must be stable under dynamical decoupling.
- **Environmental veto**: no dominant correlation with temperature, vibrations, or RF noise.
- **Null controls**: segments with no variation in $\Delta \ln \Phi$ must yield $\beta = 0 \pm \sigma$.

A control failure classifies the result as a non-interpretable systematic.

7.7. Epistemic status

A positive result at stage 03 does not constitute a standalone proof. It acts as an **orthogonal confirmatory channel**:

- coherent with clocks $\Rightarrow$ strengthening.
- incoherent $\Rightarrow$ probable instrumental-origin signal.

No cross-compensation is allowed: a quantum success does not rescue a clock failure, and conversely.

7.8. Outcome of stage 03

- **ORTHOGONAL CONFIRMATION**: possible move to stage 04.
- **NULL / LIMIT**: published constraint on $|\beta|$.
- **SYSTEMATIC**: abandonment of the $\Phi(x)$ interpretation.

This stage marks the boundary between a metrological program and truly universal physics.

8. STAGE 04 — NETWORK COHERENCE AND GLOBAL SIGNATURE

8.1. Regime change

The previous stages test local, controlled, quasi-static effects. Stage 04 targets a different regime: a **distributed signature**, observable simultaneously across multiple independent nodes.

The objective is not to increase precision, but to detect a *common structure* impossible to attribute to uncorrelated local artifacts.

8.2. Network observable

This stage defines **mathematically** the network observable and what constitutes a PASS. The objective is to test the existence of a common component $s(t)$ coherent across sites, incompatible with uncorrelated local artifacts.

Per-site data. Each site $k \in \{1, \dots, N\}$ provides a corrected time series:

$$r_k(t) \equiv y_k(t) - y_{k,\text{GR}}(t) - y_{k,\text{inst}}(t),$$

where $y_k(t) = \delta\nu_k(t)/\nu_k(t)$ is the fractional frequency, $y_{k,\text{GR}}$ the preregistered GR correction, and $y_{k,\text{inst}}$ the *preregistered* instrumental corrections (no tuning on residuals). Series are aligned to a common reference clock (UTC) and sampled on a regular grid.

Common model (latent process). We test the model:

$$r_k(t) = a_k s(t) + n_k(t),$$

where $s(t)$ is a common (network) process, a_k a per-site scale factor, and $n_k(t)$ site-specific noises (assumed uncorrelated across sites outside links).

Spectral coherence (primary network observable). In the frequency domain, for each pair (k, ℓ) , define coherence:

$$C_{k\ell}(f) = \frac{|S_{k\ell}(f)|^2}{S_{kk}(f) S_{\ell\ell}(f)}, \quad 0 \leq C_{k\ell}(f) \leq 1,$$

where $S_{k\ell}(f)$ is the cross-spectral density estimated with a fixed window and a preregistered method.

We then define a **global** (network) coherence over a band $\mathcal{B} = [f_1, f_2]$:

$$\bar{C}_{\mathcal{B}} \equiv \frac{2}{N(N-1)} \sum_{k < \ell} \frac{1}{|\mathcal{B}|} \int_{f_1}^{f_2} C_{k\ell}(f) df,$$

where $|\mathcal{B}| = f_2 - f_1$.

Loop closure (structural control). If the network admits multiple paths (redundant links), we impose a closure condition on at least one loop ($k \rightarrow \ell \rightarrow m \rightarrow k$): coherence must not depend on the path (otherwise link correlation). This control is formalized by comparing $\bar{C}_{\mathcal{B}}$ obtained under alternative paths (quantified criterion in Section 8.6).

Search band(s) (frozen). The main band is preregistered:

$$\mathcal{B} = [0.1, 10] \text{ mHz},$$

with explicit exclusion of diurnal lines and harmonics (predefined exclusion windows). Any band modification is a robustness analysis *non-decision-making*.

PASS (minimal coherence criterion). We declare a **network PASS** (signature candidate) if, and only if:

1. $\bar{C}_{\mathcal{B}} \geq C^*$, with a threshold fixed *a priori*: $C^* = 0.30$, and
2. coherence survives **surrogates** (Section 8.6): observed coherence exceeds the 99th percentile of the surrogate distribution on the same band, and
3. the result is invariant (within tolerance) under alternative link paths when available.

Otherwise, the outcome is a **bound** on the common component (Section 8.4/8.5).

Mandatory outputs.

- Figure: map of $C_{k\ell}(f)$ (heatmap) + $\bar{C}_{\mathcal{B}}$.
- Table: $\bar{C}_{\mathcal{B}}$, threshold C^* , surrogate quantiles, PASS/FAIL status per test.

8.3. Coherent spectral analysis

The analysis is carried out in the frequency domain. For each pair (k, ℓ) , compute:

$$C_{k\ell}(f) = \frac{|S_{k\ell}(f)|^2}{S_k(f)S_\ell(f)},$$

where $S_{k\ell}$ is the cross-spectral density.

- Preregistered search band: $f \in [0.1, 10]$ mHz.
- Explicit exclusion of diurnal lines and network harmonics.

A coherent and stable excess in the band constitutes a candidate signal.

8.4. Bayesian decision

This section freezes the models, priors, and robustness rule. The objective is to evaluate evidence in favor of a common (network) component without “optimizing” priors on the data.

Compared models. We compare:

- H_0 : no common component, $s(t) \equiv 0$; the $r_k(t)$ share no correlated power (aside from link noise handled by controls).
- H_1 : presence of a common component $s(t)$ such that $r_k(t) = a_k s(t) + n_k(t)$, with nonzero correlated power on band $\mathcal{B}$.

Parameter of interest (network). We parameterize the RMS amplitude of the common component on band $\mathcal{B} = [f_1, f_2]$ as:

$$A \equiv \left(\int_{f_1}^{f_2} S_c(f) df \right)^{1/2}, \quad A \geq 0,$$

where $S_c(f)$ is the spectral density of the common component.

Prior (frozen) and motivation. We adopt a **weakly informative** prior on A :

$$A \sim \text{Half-Normal}(0, \sigma_A),$$

with σ_A fixed *a priori* as a conservative “metrological” scale:

$$\sigma_A = 10^{-18} \quad (\text{in RMS fractional frequency over the band}).$$

Motivation: this prior broadly covers the orders of magnitude accessible to current clocks/networks without artificially concentrating mass at amplitudes optimized on the data. Any modification of σ_A is forbidden after seeing results (except via preregistration of a new version).

Bayes factor and threshold. The decision statistic is:

$$\ln B_{10} \equiv \ln \frac{p(\{r_k\} | H_1)}{p(\{r_k\} | H_0)}.$$

Preregistered threshold:

$$\ln B_{10} \geq 5 \quad \Rightarrow \quad \text{strong evidence for a common component on } \mathcal{B}.$$

This threshold is conservative and not optimized on the data.

Prior sensitivity (preregistered robustness). We run a robustness analysis limited to at most **three** variants, predeclared:

$$\sigma_A \in \{10^{-19}, 10^{-18}, 10^{-17}\}.$$

Robustness rule:

- a Bayesian PASS requires $\ln B_{10} \geq 5$ for the nominal value $\sigma_A = 10^{-18}$,
- and no extreme reversal (e.g. $\ln B_{10} \leq 0$) for the two alternative priors.

If the conclusion depends strongly on the prior (reversal), the result is classified as *inconclusive* and published as such.

Combined rule with frequentist coherence. The network decision follows the conservative rule (Section 3.4):

$$\text{SIGNATURE} \iff (\bar{C}_{\mathcal{B}} \geq C^*) \wedge (\ln B_{10} \geq 5) \wedge (\text{controls in Section 8.6 PASS}).$$

Otherwise, a bound is published (and/or a no-go if controls FAIL).

8.5. Physical translation

A common coherent mode implies a spatial tempo fluctuation:

$$\left\langle \left| \frac{\delta\Phi}{\Phi} \right|^2 \right\rangle_{[f_1, f_2]} = \int_{f_1}^{f_2} S_c(f) df.$$

If no threshold is crossed, a spectral bound is published:

$$\left| \frac{\delta\Phi}{\Phi} \right|_{[f_1, f_2]} < \sigma_c \quad (95\% \text{ CL}).$$

8.6. Non-negotiable controls

This section is a **blocking checklist**. If a required item is missing, not logged, or not executable, stage 04 is classified **FAIL–NO-GO**.

Checklist “if missing $\Rightarrow$ FAIL”.

1. **Traceable geodetic data.** For each site: Δh (or ΔU), method, uncertainties, applied geophysical corrections. Missing $\Rightarrow$ FAIL.
2. **Minimal environmental logs.** Temperature (multi-point), pressure, humidity, vibrations (or proxy), power/UPS status, instrument events. Missing on one site or over a significant period $\Rightarrow$ FAIL.
3. **Synchronization and timestamping.** Documented UTC alignment, bounded timestamp drift/latency, and frozen alignment rules. Missing $\Rightarrow$ FAIL.
4. **Link metadata and redundancy.** Link type per interval, events (switchover/cut), and at least one permutation/alternative path if available. If an alternative exists but is not tested $\Rightarrow$ FAIL.
5. **Surrogates (anti-correlation control).** Two mandatory surrogates:
 - *Time shift:* for each pair (k, ℓ) , apply nonzero time shifts τ and recompute $\bar{C}_{\mathcal{B}}$,

- *Phase randomization* (IAAFT or preregistered equivalent) to preserve the univariate spectrum while destroying correlation.

Missing one surrogate $\Rightarrow$ FAIL.

6. **Band stability.** Recompute metrics on $\mathcal{B} \times (1 \pm 30\%)$ (robustness analysis) and verify absence of pathological dependence. Missing $\Rightarrow$ FAIL.
7. **Exclusion of known lines (predefined).** Application of preregistered exclusion windows (diurnal, harmonics, network lines). If the exclusion is modified after the fact $\Rightarrow$ FAIL.

PASS/FAIL criteria (quantified). The controls above are evaluated as follows:

- **Surrogates PASS** if observed coherence exceeds the 99th surrogate percentile:

$$\bar{C}_{\mathcal{B}} > Q_{0.99}(\bar{C}_{\mathcal{B}}^{\text{sur}}).$$

- **Alternative paths PASS** if global coherence is compatible at 2σ :

$$\left| \bar{C}_{\mathcal{B}}^{(\text{path } 1)} - \bar{C}_{\mathcal{B}}^{(\text{path } 2)} \right| \leq 2 \text{SE}(\Delta \bar{C}_{\mathcal{B}}),$$

whenever two paths exist.

- **Band stability PASS** if the relative variation remains bounded:

$$\frac{\left| \bar{C}_{\mathcal{B}(1+0.3)} - \bar{C}_{\mathcal{B}(1-0.3)} \right|}{\bar{C}_{\mathcal{B}}} \leq 0.5,$$

(conservative preregistered threshold).

Any failure $\Rightarrow$ **FAIL–NO-GO** (undiscriminated global correlation).

Mandatory outputs (audit).

- **Checklist table:** item $\rightarrow$ present/absent, metric, threshold, PASS/FAIL.
- **Surrogates figure:** distribution $\bar{C}_{\mathcal{B}}^{\text{sur}}$ + observed value.
- **Paths figure:** comparison of metrics by path/link (if applicable).

8.7. Final epistemic status

A success at stage 04 constitutes a **network signature**, not a mere local detection.

It implies:

- spatial universality,
- instrumental independence,
- incompatibility with a local artifact.

This is the highest validation level accessible without modifying relativistic geometry.

8.8. Logical closure of the program

The CHRONON program then admits three final outcomes:

- **Global signature detected,**
- **Multi-scale constraints,**
- **Methodological no-go demonstrated.**

In all three cases, the program is complete, publishable, and intellectually closed.

BOXED NOTE — FALSIFIABILITY AND NO-GO CONDITIONS

Why this box exists

An experimental program that cannot fail is not physics. It is storytelling.

CHRONON is designed to be able to conclude “**this hypothesis is not testable in this framework**” without ambiguity, without reinterpretation, and without rhetorical dilution.

This box states the exact conditions under which the program yields a *scientific no-go*.

Definition of a no-go

We call a **CHRONON no-go** any situation in which, despite protocol-compliant execution, no valid physical conclusion about $\Phi(x)$ can be drawn.

A no-go is not a failure. It is a logical result.

Type I No-Go — Absence of a robust signal

Condition.

- All critical gates are **PASS**.
- The decision threshold is not reached: $|\hat{\varepsilon}_\Phi| < 5\sigma$.

Consequence.

$$|\varepsilon_\Phi| < 1.96 \sigma_{\hat{\varepsilon}_\Phi} \quad (95\% \text{ CL})$$

Interpretation. The field $\Phi(x)$, if it exists, produces no observable effect beyond the achieved experimental bound.

This is a physical constraint, not indecision.

Type II No-Go — Systematic failure

Condition.

- At least one critical gate fails (control, permutation, environmental veto, robustness).

Consequence.

- Any estimate of ε_Φ is disqualified.
- No physical claim is allowed.

Interpretation. The experimental setup is not able to discriminate a physical effect from an instrumental illusion.

The program stops at this level for this specific protocol.

Type III No-Go — Inter-stage inconsistency

Condition.

- A signal passes stage 01,
- fails at stage 02 (replication),
- or becomes incompatible with stage 03 (orthogonal observable).

Consequence.

- The signal is classified as *non-universal*.
- No extrapolation is allowed.

Interpretation. A real but non-universal effect is not compatible with a fundamental tempo field.

Type IV No-Go — Absence of a network signature

Condition.

- Local stages are positive or constraining,
- but total absence of inter-site coherence at stage 04.

Consequence.

- Any interpretation as a global field is excluded.
- Results remain local and contingent.

Interpretation. The observed phenomenon, if it exists, is not a distributed structure of space-time.

Closure principle

CHRONON does not allow:

- shifting thresholds after the fact,
- changing observables to rescue a signal,
- re-labeling a systematic as a “hint”.

Each outcome — signal, limit, or no-go — is final for the considered version of the protocol.

Scientific value of the no-go

A well-established no-go:

- closes a model space,

- saves decades of experimental wandering,
- clarifies what metrology can or cannot test.

In CHRONON, falsifiability is not a promise. It is a structural obligation.

FORMAL APPENDICES — NOTATION, UNITS, AND CONVENTIONS

Appendix A — Fundamental conventions

Spacetime. Coordinates:

$$x^\mu = (ct, x, y, z), \quad \partial_\mu = \left(\frac{1}{c} \partial_t, \nabla \right).$$

The relativistic metric is assumed given, fixed, and unchanged. No modification of light cones is introduced.

Gravity. The gravitational field is treated in the weak, quasi-static approximation:

$$\Delta U \simeq g \Delta h, \quad \frac{\Delta U}{c^2} \ll 1.$$

Time. Coordinate time t is that of metrological standards (UTC). Relativistic proper time is corrected according to GR, then removed from the analyzed observables.

Appendix B — Chronon field and derived quantities

Operational definition. The chronon field $\Phi(x)$ is a scalar field defining an effective local tempo, with no energy content and no metric directly measurable.

The operational relation is:

$$d\tau = \Phi^{-1}(x) dt.$$

Any global multiplicative constant of Φ is unobservable.

Measurable increment. Only spatial or temporal variations are admissible:

$$\Delta \ln \Phi \equiv \ln \Phi(B) - \ln \Phi(A) \simeq \frac{g \Delta h}{c^2}.$$

Appendix C — Observables and symbols

Symbol	Definition	Unit
y	Fractional frequency $\delta\nu/\nu$	1
y_{mes}	Measured frequency	1
y_{GR}	Relativistic prediction	1
Y	Residual $y_{\text{mes}} - y_{\text{GR}}$	1
Δh	Height difference (high – low)	m
X	$g\Delta h/c^2$	1
ε_Φ	Signed residual slope	1
α	Instrumental intercept	1
T_1	Relaxation time	s
T_2	Coherence time	s
T_ϕ	Pure dephasing	s
β	Quantum-coherence coupling	s ⁻¹

Appendix D — Signs and orientations

Sign convention.

- $\Delta h > 0$ corresponds to an elevation.
- $X = g\Delta h/c^2 > 0$.
- A slope $\varepsilon_\Phi > 0$ means a temporal residual increasing with altitude.

Any inversion of convention is explicitly logged and tested by permutation.

Appendix E — Units and constants

- Local gravitational acceleration: $g = 9.81 \text{ m s}^{-2}$ (documented local value).
- Speed of light: $c = 299\,792\,458 \text{ m s}^{-1}$ (exact).
- Useful factor:

$$\frac{g}{c^2} \simeq 1.09 \times 10^{-16} \text{ m}^{-1}.$$

All quantities are expressed in SI units.

Appendix F — Explicit assumptions

The CHRONON framework assumes:

- full validity of metric General Relativity,
- no measurable energetic backreaction of Φ ,
- strictly differential observability,
- local linearity in the weak regime.

Any violation of these assumptions constitutes a change of theory, not an extension of the present protocol.

Appendix G — What the protocol does not claim

CHRONON does not claim to:

- define the ontological origin of time,
- modify relativistic causality,
- introduce a new fundamental interaction,
- extrapolate to the cosmological regime.

The protocol tests a *measurable structure*, not a metaphysics.

Formal closure

These appendices freeze the language. Any later result is interpreted *only* within this framework.

Changing a convention means changing the experiment.

REFERENCES

- [1] T. E. Mehlstäubler, G. Grosche, C. Lisdat, P. O. Schmidt, and H. Denker, “Atomic clocks for geodesy,” *Reports on Progress in Physics* **81**(6), 064401 (2018). doi: 10.1088/1361-6633/aab409.
- [2] J. Grotti *et al.*, “Geodesy and metrology with a transportable optical clock,” *Nature Physics* **14**, 437–441 (2018). doi: 10.1038/s41567-017-0042-3.
- [3] T. Bothwell *et al.*, “Resolving the gravitational redshift across a millimetre-scale atomic sample,” *Nature* **602**, 420–424 (2022). doi: 10.1038/s41586-021-04349-7.
- [4] O. Lopez *et al.*, “Frequency and time transfer for metrology and beyond using telecommunication network fibres,” *Comptes Rendus Physique* **16**, 531–539 (2015). doi: 10.1016/j.crhy.2015.04.005.
- [5] S. Droste *et al.*, “Optical-frequency transfer over a single-span 1840 km fiber link,” *Physical Review Letters* **111**, 110801 (2013). doi: 10.1103/PhysRevLett.111.110801.
- [6] G. Grosche *et al.*, “Optical frequency transfer via 146 km fiber link with 10^{-19} relative accuracy,” *Optics Letters* **34**(15), 2270–2272 (2009). doi: 10.1364/OL.34.002270.
- [7] G. Petit and B. Luzum (eds.), *IERS Conventions (2010)*, IERS Technical Note No. 36 (2010). (International Earth Rotation and Reference Systems Service).
- [8] W. K. Newey and K. D. West, “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix,” *Econometrica* **55**(3), 703–708 (1987). doi: 10.2307/1913610.
- [9] D. W. K. Andrews, “Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation,” *Econometrica* **59**(3), 817–858 (1991). doi: 10.2307/2938229.
- [10] P. J. Huber, “Robust Estimation of a Location Parameter,” *Annals of Mathematical Statistics* **35**(1), 73–101 (1964). doi: 10.1214/aoms/1177703732.
- [11] D. York, N. M. Evensen, M. López-Martínez, and J. De Basabe Delgado, “Unified equations for the slope, intercept, and standard errors of the best straight line,” *American Journal of Physics* **72**(3), 367–375 (2004). doi: 10.1119/1.1632486.
- [12] W. E. Deming, *Statistical Adjustment of Data*, John Wiley & Sons, New York (1943). (Dover reprint ed. exists).
- [13] R. E. Kass and A. E. Raftery, “Bayes Factors,” *Journal of the American Statistical Association* **90**(430), 773–795 (1995). doi: 10.1080/01621459.1995.10476572.
- [14] H. Jeffreys, *Theory of Probability*, 3rd ed., Oxford University Press (1961).
- [15] P. D. Welch, “The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms,” *IEEE Transactions on Audio and Electroacoustics* **15**(2), 70–73 (1967).
- [16] J. S. Bendat and A. G. Piersol, *Random Data: Analysis and Measurement Procedures*, 4th ed., Wiley (2010).
- [17] T. Schreiber and A. Schmitz, “Improved Surrogate Data for Nonlinearity Tests,” *Physical Review Letters* **77**(4), 635–638 (1996). doi: 10.1103/PhysRevLett.77.635.
- [18] G. Lancaster *et al.*, “Surrogate data for hypothesis testing of physical systems,” *Physics Reports* **748**, 1–60 (2018). doi: 10.1016/j.physrep.2018.06.001.

